



ST BENEDICT'S

SUBJECT	Mathematics	PAPER	2
GRADE	11	DATE	19 July 2024
EXAMINER	Ms Brown	MARKS	130
YOUR NAME	Memo.	MODERATOR	Gr 11 Educators
TEACHER NAME		DURATION	2 ½ Hours

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Statistics	15	28
2	Statistics	13	
3	Statistics	10	24
4	Analytical Geometry	14	
5	Analytical Geometry	16	33
6	Trigonometry	17	
7	Trigonometry	9	23
8	Euclidean Geometry	14	
9	Euclidean Geometry	14	22
10	Measurement	8	
TOTAL		130	

Grant: Qn 7+8

Me: Qn 5+6

Shana: Qn 9+10

Monique: Qn 1+2

Theresa: Qn 3+4

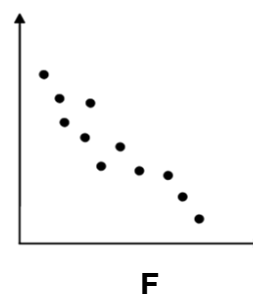
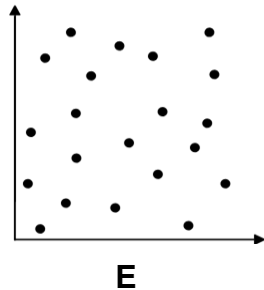
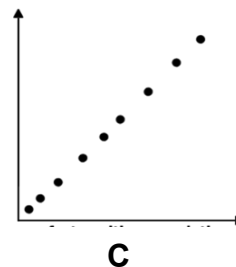
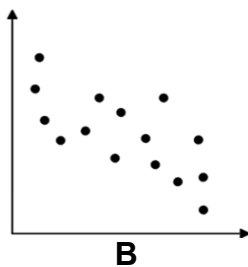
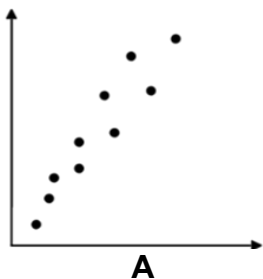
INSTRUCTIONS:

1. This paper consists of 10 Questions on 20 pages and an Information Sheet of 2 pages. Please check that your question paper is complete.
2. **Page 17** as well as a portion of **Page 13** can be used for additional working, if necessary. If this space is used, make sure that you indicate clearly which question is being answered.
3. Read the questions carefully and answer all the questions.
4. Answer all the questions on the question paper.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. All the necessary working details must be clearly shown. Answers only will not necessarily be awarded full marks.
7. It is in your interest to write legibly and to present your work neatly.
8. Round off to **ONE** decimal digit, unless otherwise stated.
9. Diagrams are not necessarily drawn to scale.

QUESTION 1

15 MARKS

- a) Match the correlation coefficients with their correct graph. Write the letter in the space given. (4)



1) $r = -0,75$ F ✓

2) $r = 0,99$ C ✓

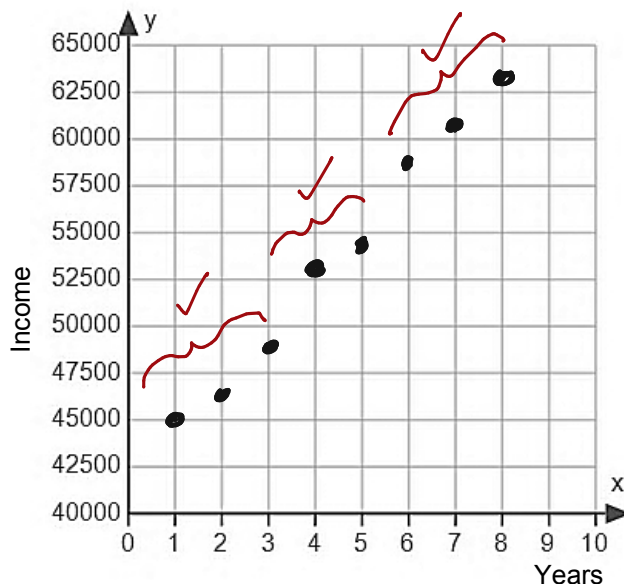
3) $r = 0,33$ D ✓

4) $r = 0$ E

b) The table below shows the income for an employee over his first 8 years of work.

Years (x)	1	2	3	4	5	6	7	8
Income (y)	45 000	46 814	48 212	52 870	54 125	58 532	61 075	62 785

1) Plot the information on the grid below: (3)



2) Determine:

i) the Linear Regression Equation to three decimal digits. (2)

$$y = 41461,536 + 2714,464x.$$

ii) the Correlation Coefficient to two decimal digits. (1)

$$0,99$$

iii) the type of correlation. (1)

Very Strong linear positive

iv) Use the linear regression equation to predict his income for his 15th year of work. Is this answer reliable? Explain your answer (4)

$$y = 41461,536 + 2714,464(15)$$

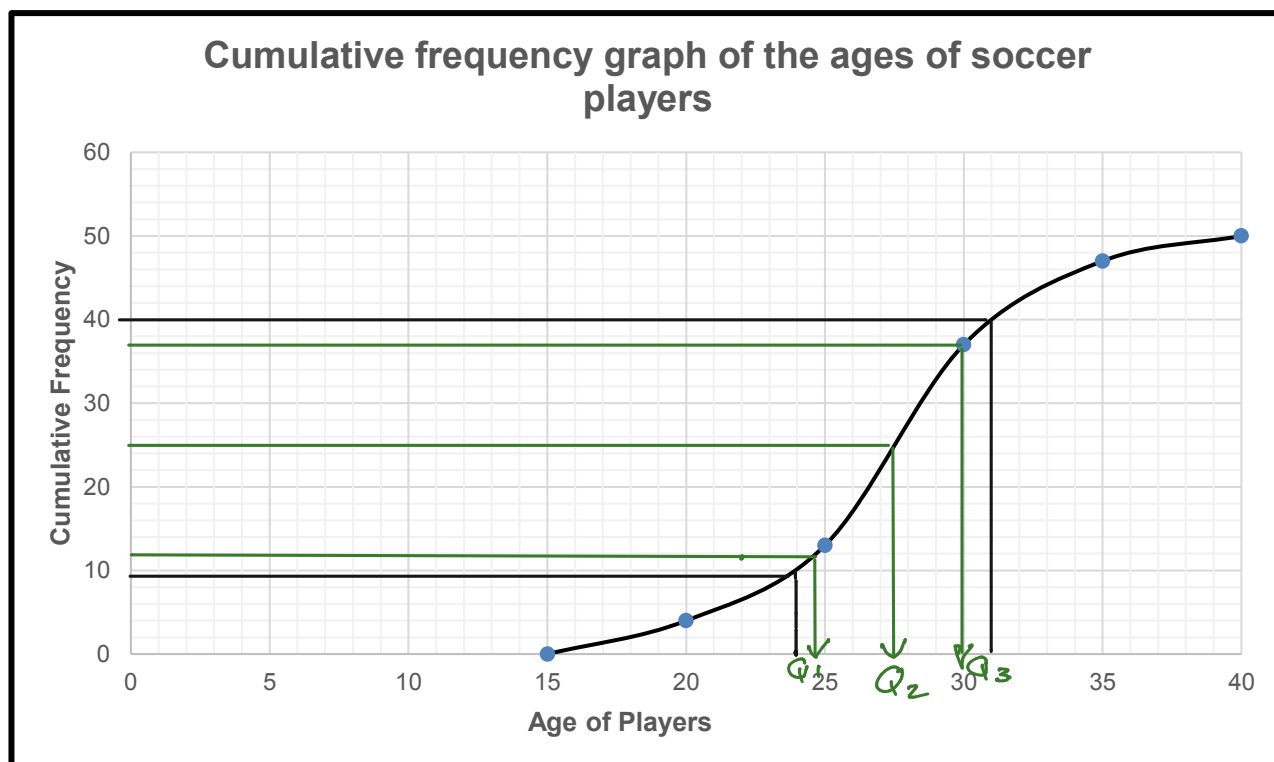
$$= 82178,50$$

No, this is not reliable. 15 is not within the data range and this is Extrapolation.

QUESTION 2

13 MARKS

A survey was conducted of the ages of players at a soccer tournament. The results are shown in the cumulative frequency graph below.



- a) How many players took part in the soccer tournament? (1)

50

✓ A.

- b) Determine the number of players between the ages of 24 and 31 years old, clearly showing where the readings are taken from on the above graph. (2)

$$40 - 10 = 30 \text{ players.}$$

✓

✓

- c) 1) Use the graph to determine the 5-number summary for the above data. (3)

$$\text{Min} = 15$$

$$Q_1 = 24.8$$

$$Q_2 = 27.5$$

$$\text{Max} = 40$$

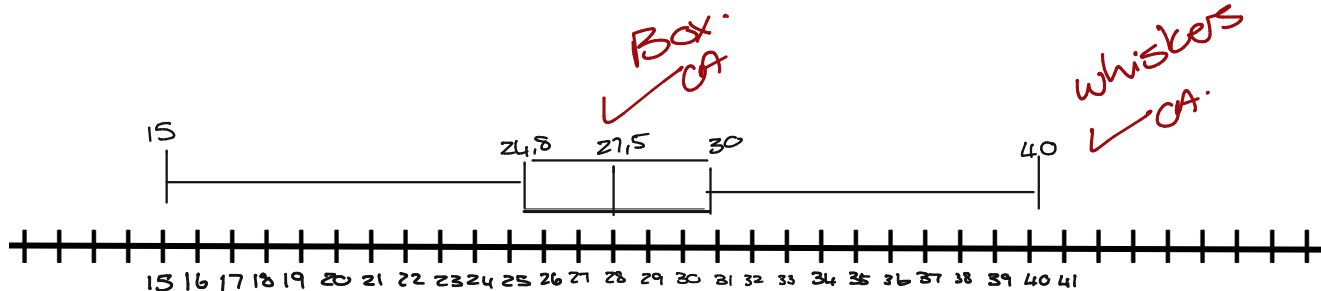
$$Q_3 = 30$$

✓

✓

✓

- 2) Draw a fully labelled box and whisker plot using the line below, as well as an appropriate scale. (2)



- d) Complete the frequency column in the table below. (2)

Class Interval	Frequency	Cumulative Frequency
$15 \leq x < 20$	4	4
$20 \leq x < 25$	9	13
$25 \leq x < 30$	24	37
$30 \leq x < 35$	10	47
$35 \leq x < 40$	3	50

- e) Calculate the Estimated Mean and describe the skewness of the data, substantiating your answer with appropriate calculations. (3)

$$\bar{X} = 27.4$$

$$\therefore \text{mean} - \text{Median} = 27.4 - 27.5 = -0.1$$

$\therefore$ Symmetrical
(or slightly skewed to the left)

QUESTION 3**10 MARKS**

The weights (in kg) of the school's top 15 rugby players are given below.

72 99 102 102 107 116 116 118 120 126 129 130 132 134 136

- a) Determine the mean weight of the rugby players. (2)

$$\bar{x} = 115,9 \quad \checkmark \quad A$$

- b) Determine the standard deviation of the above data. (1)

$$s = 16,7 \quad \checkmark \quad A$$

- c) What percentage of the players have weights that fall within one standard deviation of the mean? (2)

$$\bar{x} + s = 115,9 + 16,7 = 132,6 \quad \checkmark \quad A$$

$$\bar{x} - s = 115,9 - 16,7 = 99,2$$

$$\therefore \frac{11}{15} \times 100 = 73,3\% \quad \checkmark \quad CA$$

- d) Outliers are defined as a value less than $Q_1 - 1,5 \times IQR$ or more than $Q_3 + 1,5 \times IQR$.

Is there an outlier in the above data? Show all calculations.

(3)

$$IQR = 130 - 102 = 28$$

$$\therefore 102 - 1,5 \times 28 = 60 \quad \checkmark \text{ w}$$

$$130 + 1,5 \times 28 = 172 \quad \checkmark \text{ m}$$

$\therefore$ No outliers in the data. $\checkmark \text{ CA.}$

- e) If the coach wrote 72 for the lowest weight by mistake, and the lowest weight should have been 98 instead, how will this affect the mean and the standard deviation?

No calculations are necessary.

(2)

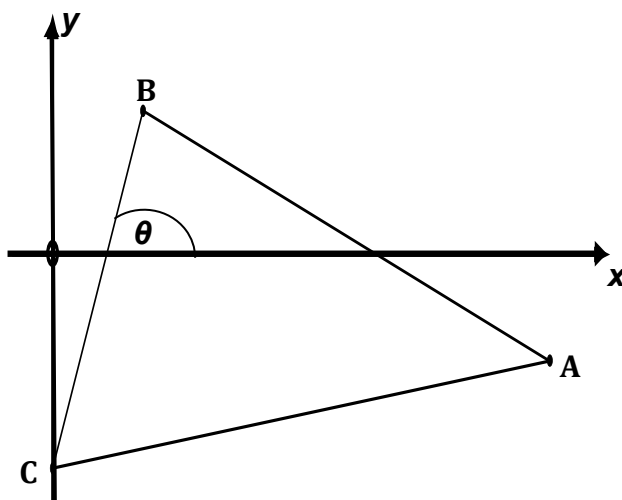
The mean would increase $\checkmark \text{ A}$

The standard deviation would decrease, $\checkmark \text{ A}$

QUESTION 4

14 MARKS

In the diagram, $A(x; y)$, $B(2; 4)$ and $C(0; -6)$ are vertices of $\triangle ABC$. The equation of the line AB is given by $y = -x + 8$.



a) Determine the:

- 1) coordinates of M, the midpoint of BC. (2)

$$M \left(\frac{2+0}{2}, \frac{4-6}{2} \right)$$

$$M(1; -1)$$

- 2) length of BC in simplest surd form. (2)

$$BC = \sqrt{(4+6)^2 + (2-0)^2}$$

$$= \sqrt{104}$$

$$= 2\sqrt{26}$$

- 3) gradient of BC. (2)

$$m_{BC} = \frac{4+6}{2-0}$$

$$= 5$$

- 4) size of θ , the inclination of line BC. (2)

$$\tan \theta = 5 \quad \checkmark \text{ m/cor.}$$
$$\theta = 78,7^\circ \quad \checkmark \text{ cor.}$$

- b) Show that the equation of the line perpendicular to BC, passing through point M, is $5y = -x - 4$ if it is given that M is the point $(1; -1)$. (2)

$$y = -\frac{1}{5}x + c \quad \checkmark \text{ gradient.}$$
$$(1; -1) \quad -1 = -\frac{1}{5} + c \quad \checkmark \text{ substitution}$$
$$c = -\frac{4}{5}$$
$$\therefore y = -\frac{1}{5}x - \frac{4}{5}.$$

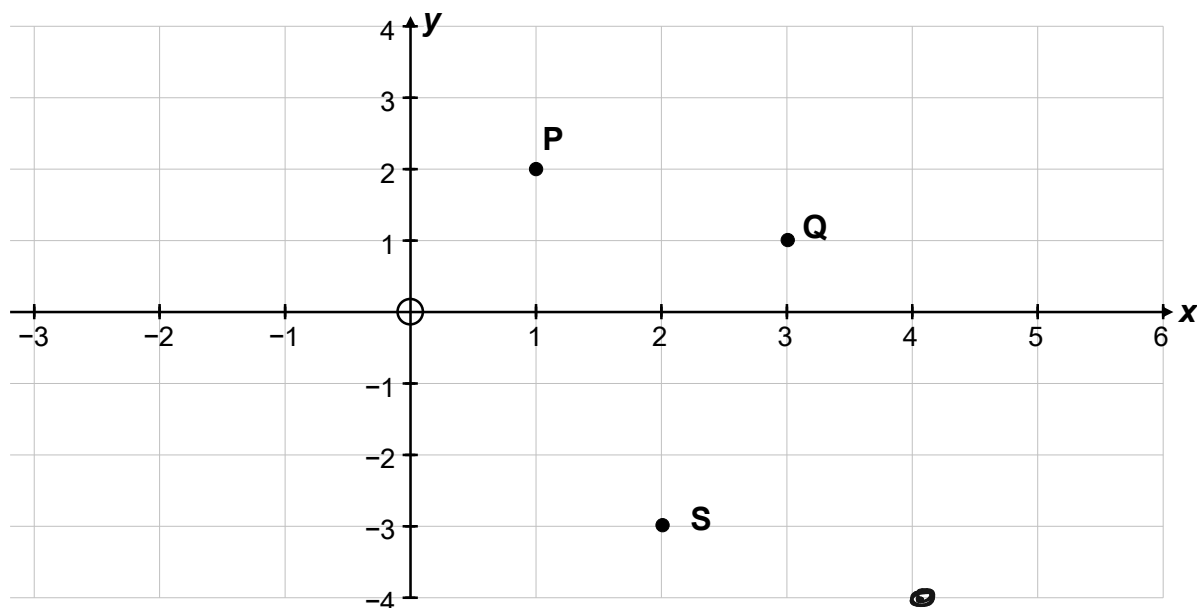
- c) It is given that $5y = -x - 4$ passes through point A. Determine the coordinates of A. (4)

$$-x + 8 = -\frac{1}{5}x - \frac{4}{5} \quad \checkmark \text{ A.}$$
$$x - 8 = \frac{1}{5}x + \frac{4}{5}$$
$$5x - 40 = x + 4 \quad \checkmark \text{ m.}$$
$$4x = 44$$
$$x = 11 \quad \checkmark \text{ A.}$$
$$\therefore y = -11 + 8$$
$$= -3 \quad \checkmark \text{ cor.}$$

QUESTION 5

16 MARKS

Consider the following points on a Cartesian plane where $P(1; 2)$; $Q(3; 1)$, $R(-3; k)$ and $S(2; -3)$.



a) Determine the value(s) of k if:

1) $T(-1; 3)$ is the midpoint of PR . (2)

$$\frac{k + 2}{2} = 3 \quad \checkmark^m$$

$$k = 4. \quad \checkmark^A$$

2) $PQ \parallel RS$. (3)

$$m_{PQ} = m_{RS}$$

$$\frac{2-1}{1-3} = \frac{k+3}{-3-2} \quad \checkmark^m$$

$$\frac{1}{-2} = \frac{k+3}{-5} \quad \checkmark^A$$

$$-5 = -2k - 6$$

$$1 = -2k.$$

$$k = -\frac{1}{2}. \quad \checkmark$$

3) $PQ \perp RS$

$$\frac{2-1}{1-3} \times \frac{k+3}{-5} = -1$$

$$\frac{1}{-2} \times \frac{k+3}{-5} = -1$$

$$\frac{k+3}{10} = -1$$

$$k+3 = -10$$

$$\therefore k = -13$$

Alt: (3)

$$m_{PQ} = -\frac{1}{2}$$

$$\therefore \frac{k+3}{-5} = 2$$

$$\therefore k+3 = -10$$

$$k = -13$$

4) P, Q and R are collinear.

$$\frac{2-1}{1-3} = \frac{1-k}{3+3}$$

$$\frac{1}{-2} = \frac{1-k}{6}$$

$$-3 = 1-k$$

$$k = 4$$

5) $RS = 5\sqrt{2}$

$$5\sqrt{2} = \sqrt{(-3-2)^2 + (k+3)^2}$$

$$50 = 25 + (k+3)^2$$

$$(k+3)^2 = 25$$

$$k+3 = \pm 5$$

$$k = 2 \quad \text{or} \quad k = -8$$

Alt:

$$\sqrt{(-3-2)^2 + (k+3)^2} = 5\sqrt{2}$$

$$25 + k^2 + 6k + 9 = 50$$

$$k^2 + 6k - 16 = 0$$

$$(k+8)(k-2) = 0$$

$$k = -8 \quad \text{or} \quad k = 2$$

b) Determine the coordinate of W if PQWS is a parallelogram.

$$P \rightarrow Q : (x+2, y-1)$$

$$S \rightarrow W : (x+2, y-1)$$

$$W (4, -4)$$

QUESTION 6

17 MARKS

a) Simplify fully:

$$\cos(180^\circ + x) \cdot \sin(90^\circ - x) + \tan x \cdot \cos x \cdot \sin(x - 180^\circ) \quad (6)$$

$$\begin{aligned} &= (-\cos x)(\cos x) + \frac{\sin x}{\cos x} \cdot \cos x \cdot (-\sin x) \\ &= -\cos^2 x - \sin^2 x \\ &= -(\cos^2 x + \sin^2 x) \\ &= -1. \end{aligned}$$

b) Simplify fully to a single trigonometric ratio, showing all working, by first reducing all angles to acute angles:

$$\frac{\sin 128^\circ \cdot \cos 232^\circ \cdot \tan 322^\circ}{\cos 412^\circ} \quad (7)$$

$$\begin{aligned} &= \frac{\sin(180^\circ - 52^\circ) \cdot \cos(180^\circ + 52^\circ) \cdot \tan(360^\circ - 38^\circ)}{\cos 412^\circ} \\ &= \frac{(\sin 52^\circ)(-\cos 52^\circ)(-\tan 38^\circ)}{\cos 52^\circ} \\ &= \frac{\sin 52^\circ}{1} \cdot \frac{\sin 38^\circ}{\cos 38^\circ} \\ &= \frac{\cos 38^\circ}{1} \cdot \frac{\sin 38^\circ}{\cos 38^\circ} \\ &= \sin 38^\circ \text{ or } \cos 52^\circ \end{aligned}$$

c) Prove the identity: $\frac{1}{\cos \theta} - \frac{\cos \theta}{1+\sin \theta} = \tan \theta$ (4)

$$\begin{aligned}
 \text{LHS} &= \frac{1+\sin \theta - \cos^2 \theta}{\cos \theta (1+\sin \theta)} \quad \text{multiply up} \\
 &= \frac{1+\sin \theta - (1-\sin^2 \theta)}{\cos \theta (1+\sin \theta)} \quad \text{change identity} \\
 &= \frac{\cancel{1} + \sin \theta - \cancel{1} + \sin^2 \theta}{\cos \theta (1+\sin \theta)} \\
 &= \frac{\sin \theta (1+\sin \theta)}{\cos \theta (1+\sin \theta)} \quad \text{factorise} \\
 &= \frac{\sin \theta}{\cos \theta} \\
 &= \tan \theta \\
 &= \text{RHS}
 \end{aligned}$$

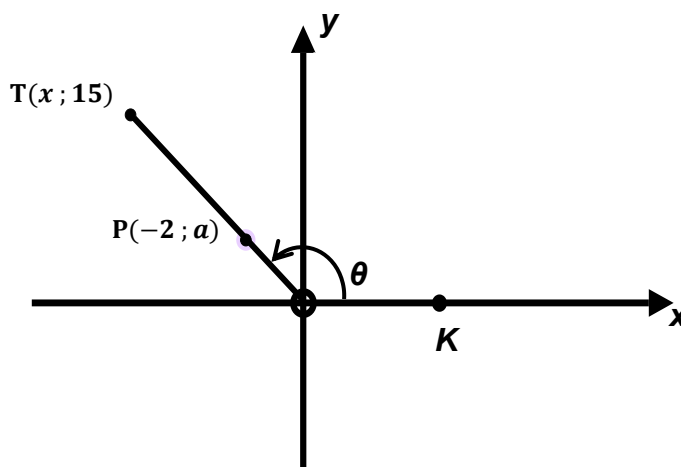
Space for Extra Working:

QUESTION 7

9 MARKS

In the diagram below, $T(x; 15)$ is a point in the Cartesian plane such that $OT = 17$ units.

$P(-2; a)$ lies on OT . K is a point on the positive x -axis and $\widehat{TOK} = \theta$.



Determine the following:

- a) the value of x . (2)

$$x^2 = 17^2 - 15^2 \quad \text{Pythag} \quad \checkmark \quad m$$

$$x = -8 \quad \checkmark \quad A$$

- b) $\tan \theta$. (1)

$$-\frac{15}{-8} \quad \checkmark \quad CA$$

- c) $\cos(\theta - 180^\circ)$. (2)

$$= -\cos \theta \quad \checkmark \quad A$$

$$= -\left(-\frac{8}{17}\right)$$

$$= \frac{8}{17} \quad \checkmark \quad CA$$

- d) the value of a .

$$\tan \theta = \frac{15}{-8} \quad \checkmark \quad CA$$

$$\theta = 118,07^\circ \quad \checkmark \quad CA$$

$$\therefore \tan 118,07^\circ = \frac{a}{-2} \quad \checkmark \quad m$$

$$\therefore a = 3,75 \quad \checkmark \quad CA$$

$$= \frac{15}{4}$$

$$\begin{array}{l} -8 : -2 \quad \checkmark \\ 15 : \frac{15}{4} \quad \checkmark \\ \therefore a = \frac{15}{4} \quad \checkmark \\ \text{ratio} \end{array}$$

$$y = mx + c \quad (4)$$

$$y = \frac{15}{-8}x \quad \checkmark$$

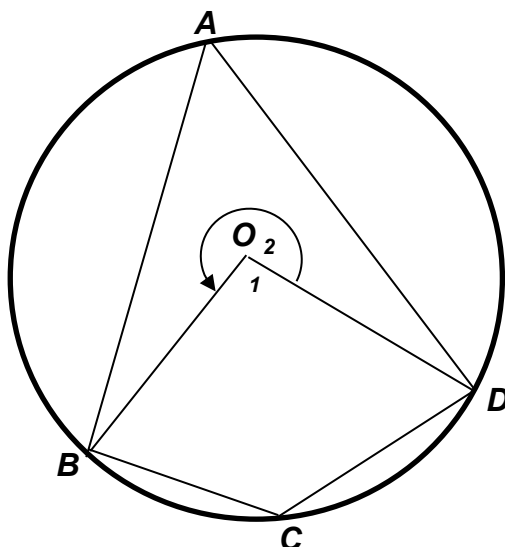
$$y = \frac{15}{-8}(-2) \quad \checkmark$$

$$= \frac{15}{4} \quad \checkmark \quad m$$

QUESTION 8

14 MARKS

- a) In the diagram below, cyclic quadrilateral ABCD is drawn in the circle with centre O.



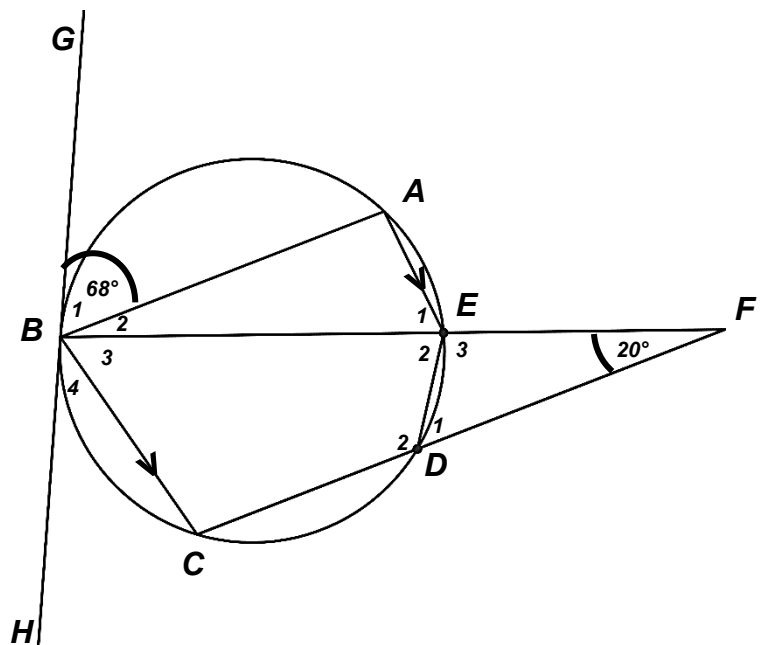
- 1) Complete the following statement by filling in the blank space: (1)

The angle subtended by a chord at the centre of the circle, is 2x the angle subtended by the same chord at the circumference of the circle.

- 2) Fill in the spaces below, using the diagram above: (5)

Statement	Reason
$\hat{O}_2 = 2\hat{A}$	$\angle$ at center = 2x $\angle$ at circumference
$\hat{O}_2 = 2\hat{C}$	$\angle$ at center = 2x $\angle$ at circumference.
$\hat{O}_1 + \hat{O}_2 = 360^\circ$	$\angle$'s round a pt.
$\therefore 2\hat{A} + 2\hat{C} = 360^\circ$	$\angle$'s round a pt
$\therefore \hat{A} + \hat{C} = 180^\circ$	

- b) In the diagram below, A, B, C, D, and E are points on the circumference of the circle such that $AE \parallel BC$. BE and CD produced meet at F. GBH is a tangent to the circle at B. $\hat{B}_1 = 68^\circ$ and $\hat{F} = 20^\circ$.



Determine the size of each of the following, giving reasons for your statements:

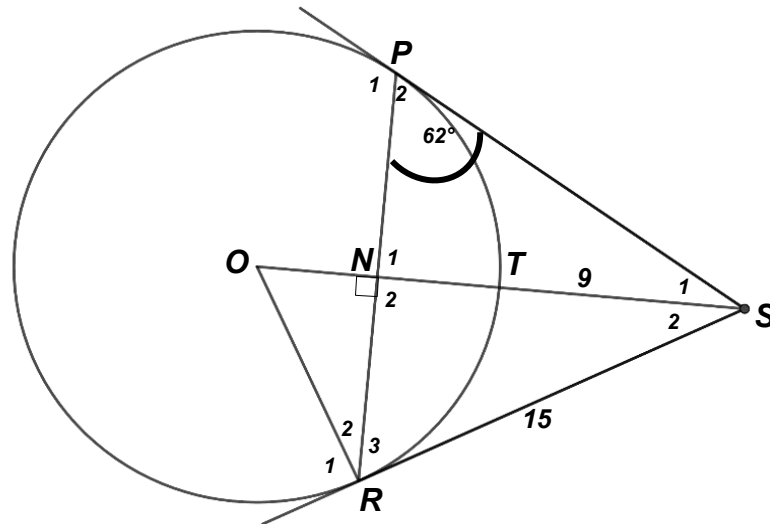
- 1) $\hat{E}_1$ (2)
 68° ✓ *tan-chord theorem* ✓ *A.*
- 2) $\hat{B}_3$ (1)
 68° ✓ *alt $\angle$ s $AE \parallel BC$* ✓ *CA.*
- 3) $\hat{D}_1$ (2)
 68° ✓ *ext $\angle$ of cyclic quad.* ✓ *A.*
- 4) $\hat{E}_2$ (1)
 88° ✓ *ext $\angle$ of Δ .* ✓ *CA.*
- 5) $\hat{C}$ (2)
 92° ✓ *opp $\angle$ s of a cyclic quad* ✓ *A.*

QUESTION 9

14 MARKS

In the diagram below, O is the centre of circle PTR . N is a point on chord PR such that $ON \perp PR$. RS and PS are tangents to the circle at R and P respectively.

$RS = 15$ units, $TS = 9$ units. $\angle RPS = 62^\circ$. It is further given that $NT : OR = 21 : 40$.



- a) Giving reasons for your statements, calculate the size of $\angle NOR$. (5)

$$\begin{aligned}
 SP &= SR \quad \text{tan from same pt.} \quad \checkmark A \\
 \therefore \hat{R}_3 &= 62^\circ \quad \text{isos \Delta.} \quad \checkmark A \\
 \hat{N}_2 &= 90^\circ \quad ON \perp RP \\
 \therefore \hat{S}_2 &= 28^\circ \quad \angle\text{'s in a } \Delta \quad \checkmark A \\
 \hat{R}_2 + \hat{R}_3 &= 90^\circ \quad \text{tan } \perp \text{ radl.} \quad \checkmark A \\
 \therefore \angle NOR &= 63^\circ \quad \checkmark A
 \end{aligned}$$

- b) Show that $ROPS$ is a cyclic quadrilateral. (2)

$$\begin{aligned}
 \hat{O} &= 62^\circ \quad \text{proven} \\
 \hat{P}_2 &= 62^\circ \quad \text{given}
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} \hat{O} &= 62^\circ \text{ proven} \\ \hat{P}_2 &= 62^\circ \text{ given} \end{aligned}} \right\} \checkmark EA$$

$\therefore ROPS$ is a cyclic quad
 (equal $\angle$'s subtended by same line)
 or converse $\angle$'s in same seg. $\checkmark A$

c) Determine the length of:

1) the radius of the circle (Hint: let $OR = x$).

$$(x+9)^2 = x^2 + 15^2 \text{ Pythag.}$$

$$x^2 + 18x + 81 = x^2 + 225$$

$$18x = 144$$

$$x = 8$$

$$\hat{R}_2 + \hat{R}_3 = 90^\circ \text{ tan } \perp \text{ rad.} \quad (3)$$

$$\tan 62^\circ = \frac{15}{x}$$

$$\therefore x = 7,98 \text{ units}$$

2) PR

$$\frac{NT}{OR} = \frac{21}{40}$$

$$\therefore NT = \frac{21}{40} x = \frac{21}{40} \times 8 = \frac{21}{5}$$

$$\therefore ON = \frac{19}{5}$$

$$RN^2 = 8^2 - \left(\frac{19}{5}\right)^2$$

$$= 1239$$

$$\frac{25}{25}$$

$$= 7,04$$

$$\therefore PR = 14,1 \text{ units line from center } \perp \text{ to chord.}$$

$$\frac{NT}{OR} = \frac{21}{40}$$

$$NT = \frac{21}{40} \times 8 = \frac{21}{5}$$

$$\therefore NS = 13,2 \text{ units}$$

$$\therefore NR^2 = 15^2 - 13,2^2$$

$$NR = 7,12 \text{ units}$$

$$\sin 62^\circ = \frac{NR}{OR}$$

$$NR = 8 \sin 62^\circ$$

$$= 7,06$$

$$\therefore PR = 14,12 \text{ units}$$

line from center

$\perp$ to chord

$$\sin 28^\circ = \frac{NR}{15}$$

$$NR = 15 \sin 28^\circ$$

$$= 7,04 \text{ units}$$

$$\therefore PR = 14,08 \text{ units}$$

line from center

$\perp$ to chord.

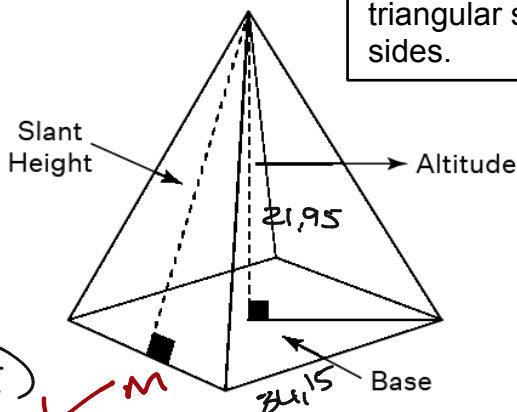
$$\therefore PR = 14,24 \text{ units}$$

line from center $\perp$ to chord.

QUESTION 10

8 MARKS

The glass pyramid at the entrance to the Louvre in Paris has an altitude height of 21,95 metres and a square base that is 34,15 metres in length.



$$\text{Volume of a pyramid} = \frac{1}{3} AH$$

Surface area of a pyramid
= area of base + area of
triangular sides $\times$ number of
sides.

- a) Find the volume of the pyramid.

$$V = \frac{1}{3} (34,15)^2 (21,95)$$

$$= 8532,86 \text{ m}^3$$

(2)

- b) Find the slant height of the side of the pyramid.

$$sl^2 = \left(\frac{34,15}{2}\right)^2 + (21,95)^2$$

$$sl = 27,81 \text{ m}$$

(3)

- c) One bottle of Windowlens cleans 15 m^2 of glass. How many bottles of Windowlens would you need to clean the outer area of the pyramid?

(3)

$$S.A = 4 \times \frac{1}{2} (34,15)(27,81)$$

$$= 1899,38 \text{ m}^2$$

$$\text{No of bottles} = \frac{1899,38}{15}$$

$$= 126,6 \text{ bottles}$$

$$\therefore 127 \text{ bottles}$$